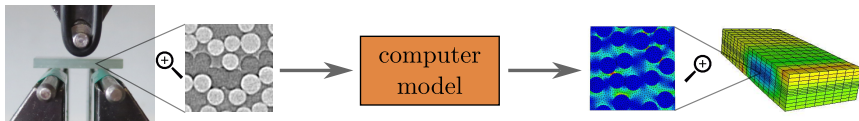


Materials and computer models – avenues for progress

The answer: combining a handful of experiments with extensive **virtual testing**

Multiscale modeling

- Model materials at very small and very large scales at the same time
- More knowledge $\Rightarrow$ smaller safety factors $\Rightarrow$ **more efficient designs**



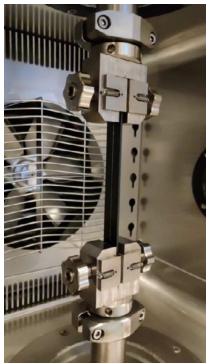
The challenge: multiscale modeling requires an immense amount of microscale computations

What we are now able to do: FE² versus experiments

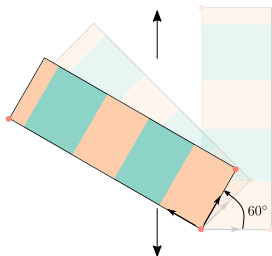
We also used it to confirm some experimental results that had been puzzling us:

- Uniaxial tension on off-axis carbon-PEEK coupons

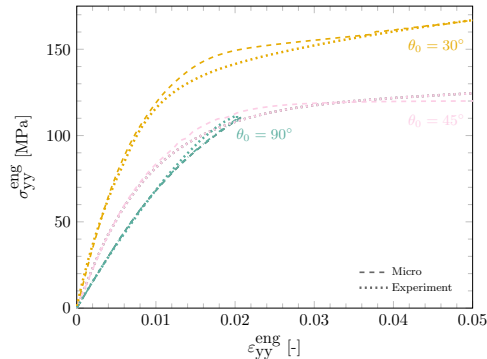
Uniaxial tension experiment



Off-axis angle



Engineering stress-strain

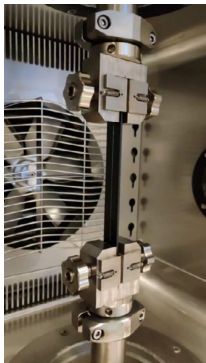


What we are now able to do: FE² versus experiments

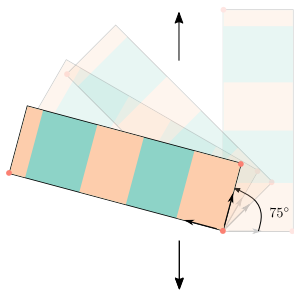
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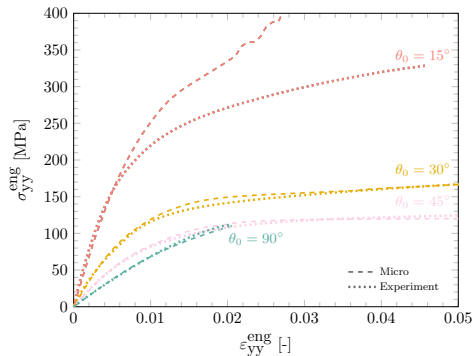
Uniaxial tension experiment



Off-axis angle



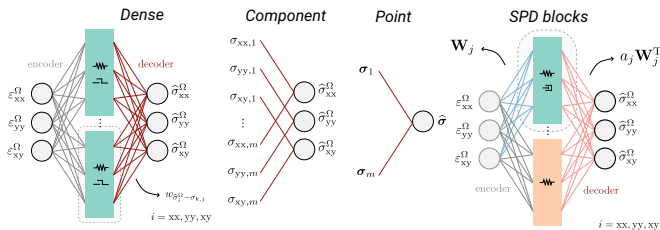
Engineering stress-strain



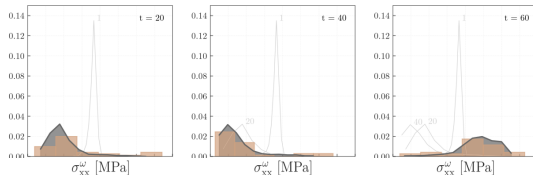
But wait, what about thermodynamics!?

Architecture flexibility allows for a lot of experimentation:

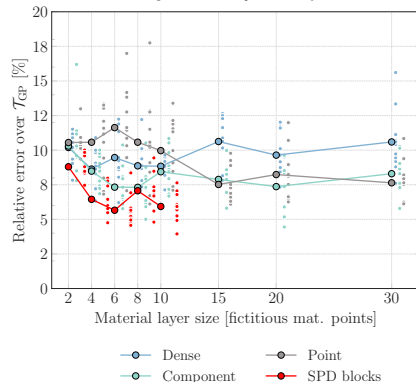
- Sparse decoder connectivities improve interpretability and local stress distributions
- SPD encoder/decoders with shared weights guarantee thermodynamic consistency



Stress histograms for a random GP path (Component architecture)



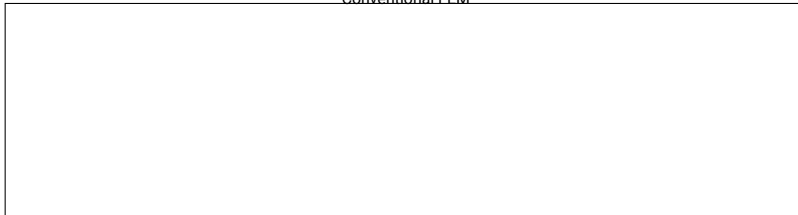
Training with only 5 GP paths



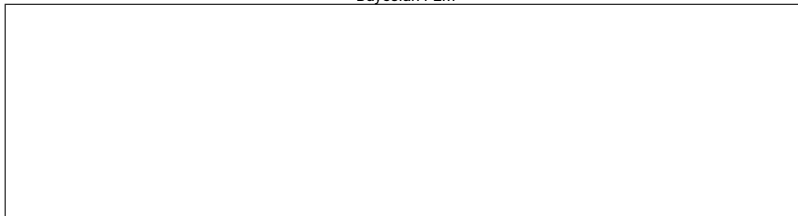
Other recent works in the lab: Bayesian FEM

Assume a prior over the solution on a fine mesh, condition on shape functions from a coarse mesh:

Conventional FEM



Bayesian FEM



[Poot, Kerfriden, Rocha, van der Meer (2025), [arXiv:2506.02815](#)]